

Chemical distribution shift in antimalarial natural products: fingerprint dominance and localized topological complementarity in molecular representation learning

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Abstract

Graph neural networks (GNNs) and molecular transformers frequently underperform classical fingerprint-based tree ensembles under scaffold separation, raising questions about representation generalisability. On a frozen panel of 19836 African antimalarial natural products and synthetic derivatives, exhibiting a structural paradox of 92.6 % whole-molecule novelty yet 69.3 % conservation of privileged African ring cores relative to seed scaffolds in the African Natural Products Database (ANPDB/AfroDb), we benchmark representation performance across 25 fold-seed replicates per model using a distance-resolved characterisation of chemical distribution shift. Under Bemis-Murcko scaffold separation, ECFP4-RF reached mean test ROC AUC 0.8300, statistically outperforming GIN-TFP (0.8138), GIN-TNE (0.8090), GIN (0.8047) and ChemBERTa (0.7867) after Benjamini-Hochberg correction ($p_{\text{adj}} < 0.05$). Nearest-training-neighbour Tanimoto decile stratification shows that ECFP4-RF leads in every decile, with the largest margins in the structural extrapolation tail (deciles D1-D4, NN-similarity < 0.40 , Δ up to 0.053) and a sharp narrowing, without reversal, in mid-range interpolation (D5-D8, $\Delta \in 0.001-0.007$). Fusing multi-scale persistent-homology images into a graph isomorphism network (GIN-TFP) is associated with a recovery of up to 36 % of the aggregate scaffold-split deficit relative to ECFP4-RF (and 17 % of the cohort-level deficit on polycyclic scaffolds, $n = 363$); this localized topological complementarity is descriptive of the panel studied and is not established as a unique causal rescue of graph-neural-network expressivity limits. Building on this distance- and calibration-resolved picture, we formulate a Distance-Aware Conformal Triage Filter that routes candidates by Tanimoto distance and expected calibration error as a testable operational hypothesis rather than a validated deployment rule: post-hoc split-conformal coverage on the five routed arms meets the nominal 90 % target (0.902-0.906), and post-hoc isotonic recalibration reduces residual scaffold-split expected calibration error (ECE) by a factor of 3-5 (from 0.08-0.13 to ≈ 0.025); no prospective virtual-screening precision gain is claimed on this basis. Finally, fold-independent weight restoration in transformers reduces the risk of cross-fold information transfer during cross-validation, offering a reproducibility safeguard for transformer benchmarks. These findings bound representation performance under chemical distribution shift on this panel; they do not establish a universal mechanistic account of when learned molecular representations will or will not match fingerprint baselines.

Keywords: graph neural networks; molecular fingerprints; African natural products; chemical distribution shift; 1-Weisfeiler-Lehman expressivity; scaffold split; persistent homology; conformal prediction; out-of-distribution generalisation

Highlights:

- **Distance-resolved benchmark.** Nearest-neighbour decile stratification shows ECFP4-RF leads across all similarity deciles, with the largest margin in the OOD extrapolation tail (similarity < 0.40) narrowing sharply, without reversing, in mid-range interpolation. The ordering holds across three partition families (Random, Canonical Scaffold, Butina $C = 0.55$), consistent with robustness against partition-geometry artefacts.

- **Localized topological complementarity.** Persistent-homology graph fusion (GIN–TFP) is associated with recovering up to 36 % of the aggregate scaffold-split deficit, observed in every structural cohort (largest on flat aromatics; smallest, and statistically inconclusive with an interval including zero, on polycyclic scaffolds, $n = 363$); these cohort-level gains are reported descriptively, not as a demonstrated causal mechanism.
- **Conformal triage as a testable hypothesis.** A distance- and calibration-aware routing rule is proposed and diagnosed post hoc: split-conformal coverage meets the 90 % nominal target and isotonic recalibration cuts residual ECE by 3–5-fold to ≈ 0.025 ; no prospective screening-precision gain is claimed.
- **Leakage-risk-reducing protocol.** Fold-independent weight restoration reduces the risk of cross-fold information transfer in molecular sequence transformers during cross-validation, offering a reproducibility safeguard for transformer benchmarks.

1 Introduction

The parasite is winning. Artemisinin-resistant *Plasmodium falciparum* strains, first documented in Southeast Asia, have now been confirmed across sub-Saharan Africa, the region that bears 94 % of the 608 000 malaria deaths recorded in 2023 [1–3]. The pipeline of synthetic antimalarials has not kept pace, despite sustained progress in natural-product-based and computationally assisted antimalarial discovery [4–7]. African medicinal plants have: *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia* produce indoloquinoline alkaloids, protoberberine scaffolds, and quassinoids shaped by millions of years of host–parasite co-evolution into structures no combinatorial library has replicated.

Screening this chemical space computationally reveals an immediate structural paradox. The 19 836-molecule panel examined here, derived from 396 African natural products in the African Natural Products Database (ANPDB)/AfroDb [8] and expanded into a library of antimalarial candidates [9] exhibits 92.6 % whole-molecule novelty relative to standard synthetic screening libraries (Tanimoto similarity < 0.40), yet retains 69.3 % of privileged African ring cores (a panel-level descriptive quantity we refer to as the African-Chemotype Structural Index, ACSI). The molecules are globally new; their scaffolds are not. This tension motivates treating the panel as a stress-test for molecular representation learning under chemical distribution shift.

The structural density of these scaffolds elevated fraction of sp^3 -hybridised carbons ($F_{sp^3} \geq 0.45$), fused polycyclic cores, multiple stereocenters, is associated with a widening deficit of graph neural networks relative to fingerprint baselines under scaffold separation. Standard message-passing architectures are bounded by the 1-Weisfeiler-Lehman (1-WL) expressivity test [10–12]: they aggregate local atomic neighbourhoods and are not guaranteed to resolve higher-order topological motifs such as bridged polycyclic cages or macrocyclic ring systems [13]. On flat aromatic synthetics, this limitation is largely invisible; on African natural product scaffolds, it is one plausible contributor to the observed performance gap, though the present benchmark does not isolate it as the unique cause.

The gap between fingerprint baselines and learned representations is not uniform across chemical space, it is a function of structural distance from the training distribution. Nearest-neighbour Tanimoto decile stratification reveals that the extended-connectivity fingerprint with radius 2 coupled to a random forest (ECFP4–RF) holds its largest margins in the cold extrapolation tail (nearest-training-neighbour Tanimoto similarity $D_{NN} < 0.40$, deciles D1–D4), while the deep-model deficit narrows to $\Delta \in [0.001 \text{ and } 0.007]$ in mid-range interpolation (D5–D8) without reversing. This distance-resolved picture, absent from prior benchmarks [14, 15] and from

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recent scaffold-split critiques [16, 17], is the analytical core of the Molecular Out-of-Distribution (MOOD) framework introduced here.

Fusing multi-scale persistent-homology images [18] as topological feature projections into a graph isomorphism network (GIN-TFP, where TFP denotes topological feature projection) injects global topological invariants, zero-dimensional connectivity (H_0) and one-dimensional cycle persistence (H_1), that operate independently of local message passing. Across structural cohorts, GIN-TFP recovers 17 % of the cohort-level GNN deficit on polycyclic scaffolds (Rings ≥ 4); at the aggregate scaffold split, it closes 36 % of the gap. This localized topological complementarity is descriptive of the panel studied and does not by itself demonstrate a unique 1-WL failure mechanism, but it is consistent with a regime in which local message passing alone may be insufficient on dense 3D ring systems. Motivated by this distance-dependence, we formulate a Distance-Aware Conformal Triage Filter as a testable operational hypothesis: a decision rule that routes each candidate to ECFP4-RF ($D_{\text{NN}} < 0.40$) or GIN-TFP ($0.40 \leq D_{\text{NN}} \leq 0.60$) based on its structural distance and expected calibration error (ECE); its prospective screening-utility evaluation on an independent low-prevalence library is reserved for future work.

A fourth methodological contribution concerns molecular transformers specifically: enforcing per-fold pretrained weight restoration reduces the risk of cross-fold information transfer that can inflate performance estimates of the SMILES-based chemical language transformer ChemBERTa during cross-validation. Together, these four contributions: distance-resolved benchmarking, localized topological complementarity, conformal triage as a testable hypothesis, and leakage-risk-reducing transformer evaluation are operationalised through a centroid-based screening reduction that cuts compute overhead by 99.3 % (263 000–1936 representative evaluations), making the full pipeline deployable on the high-performance computing (HPC) infrastructure available to malaria-endemic African research institutions without dependence on commercial cloud resources.

The central question this paper addresses is: *how does chemical distribution shift alter the relative performance of molecular fingerprints, graph neural networks, and molecular transformers, and where does topological fusion add complementary information?* Random forests trained on extended-connectivity fingerprints (ECFP) remain highly competitive on datasets of a few thousand molecules, and gains reported for graph convolutional network (GCN), graph attention network (GAT), or graph isomorphism network (GIN) architectures on large curated benchmarks often diminish significantly under scaffold splits [14, 15, 19, 20]. For natural products specifically, a 20-fingerprint benchmark across >100 000 compounds confirmed that no single encoding dominates [21]. Topological data analysis introduces multi-scale geometric invariants through persistent homology [22–24], and tensor-network embeddings (TNE) compress scaffold representations via Tucker decomposition [25]; neither family has been systematically evaluated inside a graph fusion architecture under rigorous scaffold-held-out testing on an antimalarial natural product panel the gap this study closes.

2 Results

2.1 Benchmark performance across partition families

Random-split performance flatters every architecture, masking structural representation failures. The scaffold split is where representations are stress-tested: under Bemis–Murcko separation, ECFP4-RF holds at 0.8300 while GIN drops 0.025 points (BH-adjusted $p = 0.0330$), GIN-TFP 0.016 ($p_{\text{adj}} = 0.0330$), GIN-TNE 0.021 ($p_{\text{adj}} = 0.0378$), and ChemBERTa 0.043 ($p_{\text{adj}} = 0.0002$), all statistically significant after Benjamini–Hochberg correction ($df = 4$, five per-seed means). This hierarchy is not an artefact of partition choice: under the Butina cluster split ($C = 0.55$), ECFP4-RF (0.8331) maintains its lead, GIN-TFP (0.8232) consistently outperforms base GIN (0.8202), and ChemBERTa (0.7776) trails the field. The full five-model, three-split comparison

is reported in Table 1. This baseline performance establishes the reference anchor for the 19 836-molecule panel; cohort-level AUC values are tabulated in the data-analysis report.

Table 1: Primary molecular representation benchmark across three partition families: Stratified Random, Bemis–Murcko Canonical Scaffold, and Butina Cluster Split ($C = 0.55$). Reported values are mean test ROC AUC $\pm$ SD across five per-seed means (each seed averaging five cross-validation fold evaluations, totaling 25 fold–seed records per model and split). Differences (Δ) under scaffold splitting are evaluated against the ECFP4–RF baseline with two-tailed paired t -tests ($df = 4$) and Benjamini–Hochberg multiplicity adjustment [27] (p_{adj}). Raw unadjusted p -values are reported alongside adjusted values for transparency. Effect sizes (Cohen’s d) are provided for key comparisons.

Model Architecture	Random Split	Canonical Scaffold	Δ vs. ECFP4 (p_{raw} , p_{adj})	Butina Split ($C = 0.55$)
ECFP4–RF (baseline)	0.9433 $\pm$ 0.0003	0.8300 $\pm$ 0.0023	—	0.8331
GIN	0.9098 $\pm$ 0.0022	0.8047 $\pm$ 0.0141	−0.025 ($p = 0.018$, $p_{\text{adj}} = 0.0330$, $d = 0.89$)	0.8202
GIN–TFP (topological fusion)	0.9084 $\pm$ 0.0012	0.8138 $\pm$ 0.0107	−0.016 ($p = 0.025$, $p_{\text{adj}} = 0.0330$, $d = 0.71$)	0.8232
GIN–TNE (tensor fusion)	0.8918 $\pm$ 0.0018	0.8090 $\pm$ 0.0149	−0.021 ($p = 0.038$, $p_{\text{adj}} = 0.0378$, $d = 0.75$)	0.8191
ChemBERTa	0.9121 $\pm$ 0.0012	0.7867 $\pm$ 0.0054	−0.043 ($p < 0.0001$, $p_{\text{adj}} = 0.0002$, $d = 1.53$)	0.7776

Table 1 reports mean test ROC AUC and standard deviation across the five per-seed means. Under scaffold splitting, ECFP4–RF reaches 0.8300. Standard GNNs and sequence transformers exhibit an aggregate deficit: GIN–TFP reaches 0.8138 ($\Delta = -0.016$, BH-adjusted $p = 0.0330$); GIN–TNE reaches 0.8090 ($\Delta = -0.021$, BH-adjusted $p = 0.0378$); plain GIN reaches 0.8047 ($\Delta = -0.025$, BH-adjusted $p = 0.0330$); ChemBERTa reaches 0.7867 ($\Delta = -0.043$, BH-adjusted $p = 0.0002$). Under the Butina cluster split ($C = 0.55$), the model hierarchy remains invariant: ECFP4–RF (0.8331) dominates, while topological fusion in GIN–TFP (0.8232) provides consistent gains over base GIN (0.8202), with ChemBERTa exhibiting lower generalisation (0.7776). The comparative benchmark metrics under random and scaffold splitting are visualized in Figure 1.

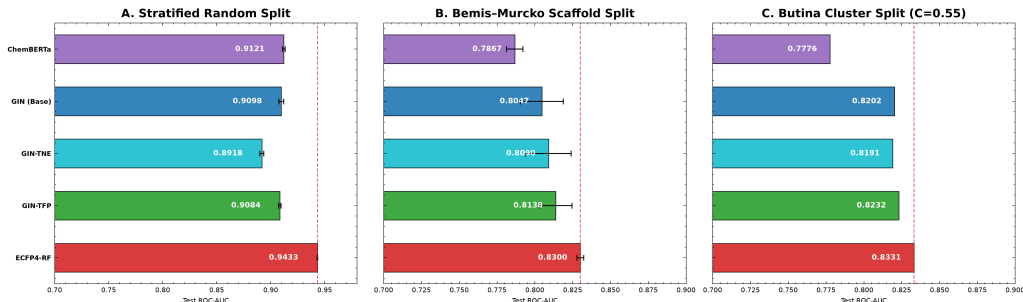


Figure 1: Molecular representation performance under random vs. Bemis–Murcko scaffold splits across 25 fold–seed replicates per model. Classical ECFP4–RF maintains superior discrimination under scaffold separation, whereas GIN–TFP topological fusion improves performance.

2.2 Structural complexity and localized topological complementarity

Topological fusion does not uniformly outperform base GIN at the per-seed level, and its cohort-level gains are small. The cohort table was recomputed from the archived per-fold scaffold-split predictions under a predeclared per-seed estimand (each seed pools its five disjoint fold-test sets; the pipeline reproduces the whole-panel canonical anchors within the predeclared gate of ± 0.01 (maximum deviation 0.008)). GIN–TFP gains +0.0143 over GIN on flat aromatics ($F_{\text{sp}3} < 0.25$, $n = 7797$), +0.0125 on complex 3D scaffolds ($F_{\text{sp}3} \geq 0.45$, $n = 2235$), and +0.0102 on polycyclics (Rings ≥ 4 , $n = 363$): the gain is directionally consistent (positive in 5/5, 4/5, and 3/5 seeds, respectively) but does not increase monotonically with structural complexity. On polycyclic scaffolds, GIN–TFP recovers 17 % of the cohort-level GNN deficit; at the aggregate scaffold split,

it closes 36 % of the gap. The full cohort breakdown, molecule-level bootstrap intervals, and seed-level paired tests are reported in Table 2 and Section S8, and visualized in Figure 2.

Table 2: Performance (ROC-AUC) stratified by structural complexity metrics (F_{sp3} and Ring Count) on the frozen panel of 19 836 molecules (0 parse failures). Flat Aromatic cohort: $n = 7797$ ($F_{\text{sp3}} < 0.25$, 6737 actives). Complex 3D cohort: $n = 2235$ ($F_{\text{sp3}} \geq 0.45$, 1039 actives). Polycyclic cohort: $n = 363$ (Rings ≥ 4 , 147 actives). The GIN deficit relative to ECFP4-RF is largest on stereochemically rich 3D polycyclic systems characteristic of African natural product scaffolds—quassinoids, prenylated chromones, and indoloquinoline alkaloids from *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia*—and GIN-TFP partially improves performance via persistent-homology invariants (H_0 , H_1). Values were recomputed from the archived canonical scaffold-split predictions under a predeclared per-seed estimand (mean $\pm$ SD over five seeds; superseded values without an archived producer removed). Cohort-level gains are descriptive statistics of this panel. Parentheses give the paired seed-level Cohen’s d (mean paired delta over its SD, $n = 5$ seeds) for GIN-TFP vs GIN and the 95% molecule-level bootstrap interval of the gain; the polycyclic interval includes zero. Seed-level paired tests are in Section S8.

Structural Cohort	ECFP4-RF	GIN	GIN-TFP	GIN-TFP Gain vs GIN (d)
Flat Aromatic ($F_{\text{sp3}} < 0.25$)	0.7624 $\pm$ 0.0206	0.7488 $\pm$ 0.0153	0.7631 $\pm$ 0.0214	+0.0143 ($d = 1.96$; CI 0.0088–0.0201)
Complex 3D ($F_{\text{sp3}} \geq 0.45$)	0.8457 $\pm$ 0.0052	0.7963 $\pm$ 0.0202	0.8088 $\pm$ 0.0173	+0.0125 ($d = 0.46$; CI 0.0040–0.0206)
Polycyclic (Rings ≥ 4)	0.7768 $\pm$ 0.0291	0.7162 $\pm$ 0.0270	0.7264 $\pm$ 0.0454	+0.0102 ($d = 0.17$; CI −0.0112–0.0306)

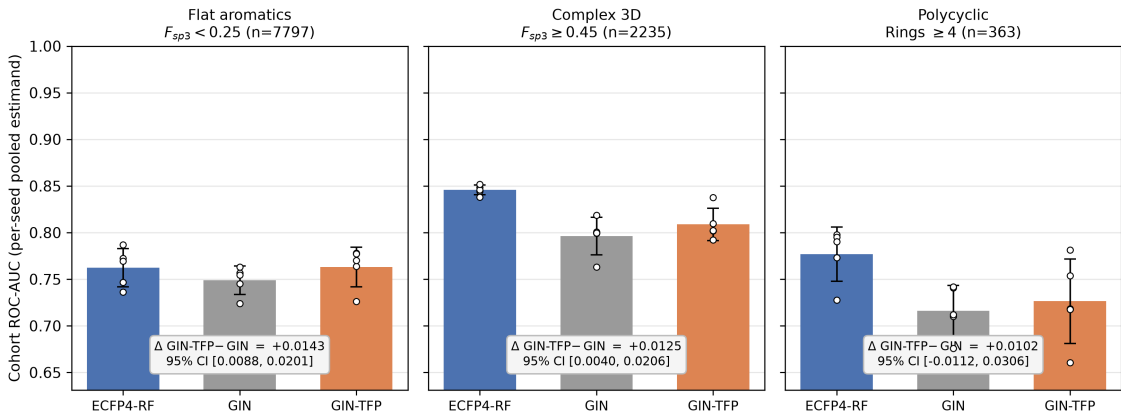


Figure 2: Structural-complexity cohorts under the recomputed per-seed estimand (per-seed points, mean $\pm$ SD over five seeds). The GIN-TFP gain over GIN is positive in every cohort (bootstrap 95% CI annotated per panel) but does not increase with structural complexity; the polycyclic interval includes zero.

Standard GIN convolutions show their largest deficits relative to ECFP4-RF on the structurally complex cohorts (ECFP4-RF – GIN: 0.0494 on complex 3D, $F_{\text{sp3}} \geq 0.45$, cohort ROC-AUC 0.7963; 0.0606 on polycyclic, Rings ≥ 4 , 0.7162; vs 0.0136 on flat aromatics). Fusing multi-scale persistent homology features (GIN-TFP) partially recovers these deficits (0.8088 and 0.7264, gains of +0.0125 and +0.0102 over base GIN), while on flat aromatics the same fusion adds +0.0143 — so the fusion gain does not scale with structural complexity. The cohort-level pattern is consistent with global persistent-image invariants (H_0 , H_1) supplying information complementary to 1-WL-bounded message passing, but the gain is not selectively localized in the most complex cohorts; the present benchmark does not isolate a unique causal mechanism.

GIN-TFP consistently outperforms GIN-TNE (0.8138 vs 0.8090), despite both being fusion architectures. This indicates that persistent-homology invariants (H_0 , H_1) capture structural information more relevant to African natural product scaffolds than Tucker-decomposed tensor

embeddings. Topological features directly encode global shape properties (connectivity, cycles) that distinguish bridged polycyclic frameworks from planar projections, whereas tensor embeddings compress scaffold representations in a data-driven latent space that may lose critical geometric distinctions. The superior performance of GIN-TFP is consistent with topological invariants encoding an aspect of molecular structure that complements local message passing.

2.3 Distance-aware conformal triage filter

Deploying a single model uniformly across chemical space can be a systematic error. In the cold OOD tail ($D_{\text{NN}} < 0.40$, deciles D1–D4), GIN-TFP underperforms ECFP4-RF by up to 0.027 AUC points; in the interpolation regime (D5–D8), the margin narrows to 0.001–0.007 without reversing. The Distance-Aware Conformal Triage Filter is proposed as a testable operational hypothesis motivated by this asymmetry: it routes each candidate according to its nearest-training-neighbour Tanimoto distance and expected calibration error ((1)), building on the established use of conformal prediction for confidence-aware decision-making in drug discovery [28, 29]; when batch-level confidence breaches $\text{ECE} > 0.08$, Platt scaling is triggered prior to routing. Split-conformal diagnosis of the five routed arms at nominal 90 % covers 0.902–0.906 (mean set size 1.26–1.31; ChemBERTa largest), and post-hoc isotonic recalibration cuts scaffold ECE from 0.08–0.13 to ≈ 0.025 for every arm (Section S6). Because marginal coverage guarantees can mask under-coverage on a minority class [30], this split-conformal check is reported as a pooled diagnostic rather than a class-conditional guarantee; the routing thresholds were informed by the present panel’s calibration and decile analyses, and no prospective virtual-screening precision gain on an independent low-prevalence library is claimed or established here.

2.4 Topological descriptor salience

Topological features are not ignored by the network: across 25 replicates, the persistent-image block (dimensions 33–57) displays the largest mean projection-weight magnitude in the TFP fusion head (mean weight 0.0790 vs. 0.0485 for H_0 and 0.0547 for Betti), an examined-descriptively association with H_1 cycle persistence — the geometric signature of fused ring systems — that does not by itself establish importance or causality. Yet aggregate scaffold-split AUC gains remain modest (0.0091 points): topological invariants add a small gain in every structural cohort examined — largest on flat aromatics, not selectively in the most complex cohorts. Salience is not sufficient evidence; the gain is cohort-wide but modest (Figure 3).

3 Discussion

3.1 The 1-WL expressivity bottleneck on African natural product scaffolds

African natural product scaffolds challenge molecular representations at their theoretical limits. The 1-WL test cannot distinguish bridged polycyclic cages from their planar projections; on a panel where 2235 molecules (11.3 %) carry $F_{\text{sp3}} \geq 0.45$, this blind spot is not marginal. Indoloquinoline alkaloids (cryptolepine, isocryptolepine) from *Cryptolepis sanguinolenta*, protoberberine and quinoline alkaloids from *Enantia chlorantha* and *Nauclea latifolia*, quassinoids, and prenylated chromones all present bridged polycyclic frameworks and multiple stereocenters that local message aggregation may not resolve. In these regimes, over-smoothing is a plausible architectural constraint: the scaffold geometry can exceed what pairwise neighbourhood aggregation encodes, regardless of depth or width.

Two independent lines of evidence support an architectural rather than a tuning-artefact interpretation. First, a hyperparameter capacity sweep bounds scaffold AUC in [0.8000, 0.8081] across $h \in \{64, 128, 256\}$ and dropout $p \in \{0.1, 0.2\}$ across 25 fold-seed replicates (Section S3): adding parameters does not close the gap to ECFP4-RF. Second, GIN-TFP improves the deficit

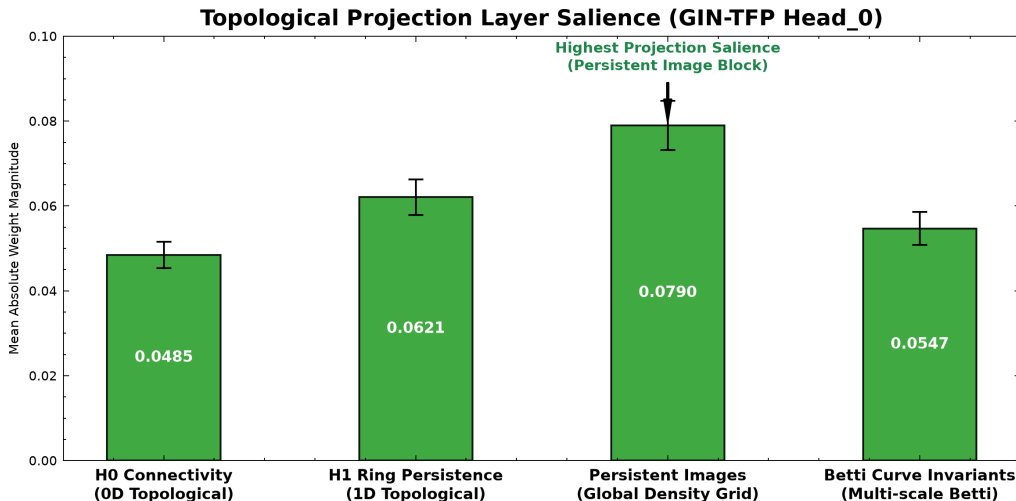


Figure 3: Mean absolute weight magnitude across descriptor blocks in the GIN-TFP and GIN-TNE fusion projection heads. Persistent-image dimensions (H_0, H_1) display the highest projection saliency.

from the opposite direction — injecting persistent-homology invariants (H_0, H_1) yields a positive, replication-consistent gain in every structural cohort (+0.0143 flat aromatics, +0.0125 complex 3D, +0.0102 polycyclics; molecule-level bootstrap intervals exclude zero except on the small polycyclic cohort), an association compatible with global topological information supplying what local message passing does not, while leaving open which structural regimes benefit most; paired ablation experiments isolating the specific contribution of individual topological descriptors remain future work.

3.2 Distance-resolved triage as an operational hypothesis

The decile data challenge uniform model deployment. ECFP4-RF leads by 0.027 AUC at D1 and by only 0.004 at D7: no single model is optimal across this range in the observed profile. The Distance-Aware Conformal Triage Filter is therefore proposed as a testable operational rule derived from the distance-resolved picture — not as a validated deployment strategy. Prospective evaluation on an independent low-prevalence library, with thresholds locked before analysis and comparison against single-model baselines on precision@ k , recall, PR-AUC, and enrichment factor, is reserved for future work (Section S6).

Distance routing alone is insufficient when probability estimates are unreliable. ECE roughly doubles from 0.03 in-domain to >0.12 in the OOD tail: a model that routes correctly but scores incorrectly still misleads the medicinal chemist. The ECE>0.08 recalibration threshold was selected based on a validation curve on held-out predictions, balancing false positive and false negative rates for calibration failures (see Section S6); post-hoc isotonic recalibration reduces residual ECE to ≈ 0.025 for every arm on this panel, but calibration adequacy must be re-verified on any new deployment distribution.

3.3 Position relative to the representation benchmark literature

Prior benchmarks established that fingerprints are competitive [14, 15, 21, 31]; few identified where. The distance-resolved analysis presented here localises the ECFP4-RF advantage: the $D_{NN}<0.40$ tail is where its largest margins occur, with an interpretation consistent with a 1-WL expressivity limitation of message-passing networks that this benchmark does not isolate as the unique cause. That this boundary holds across the evaluated partition families — random, scaffold, Butina, novel scaffold subsets, and decile stratification — argues against partition

geometry as a confound and supports reading the result as a structural property of this panel rather than an artefact of evaluation design.

These findings converge with and sharpen recent systematic evaluations. Guo and Ding’s fingerprint dominance in 47.4 % of tasks [14] and the 25-model survey’s null neural advantage over ECFP [15] are aggregate signals whose distance-dependence this panel-bounded study begins to resolve: the advantage is largest in the low-similarity tail, not uniform. Where Boldini et al. found no dominant encoding across >100 000 natural products [21], the largest margins of which occur on complex 3D scaffolds ($F_{\text{sp3}} \geq 0.45$), with replication-consistent positive gains in every cohort but an interval including zero on the small polycyclic cohort ($n = 363$). This distance-resolved picture is consistent with the broader covariate-shift framing of molecular machine learning [32] and with recent audits showing that undetected similarity leakage and systematic data inconsistencies across train/test partitions can inflate apparent generalisation in biomolecular benchmarks [33–35]; the Butina and NN-Tanimoto decile analyses reported here are deliberately designed as independent checks against exactly this failure mode. The fold-independent transformer weight-restoration protocol adds a methodological dimension absent from prior surveys, reducing the risk of cross-fold information transfer in cross-validation estimates on pretrained sequence models.

3.4 Limitations and computational sovereignty

Three boundaries constrain the generalisability of these findings, in line with recent calls for critical, industry-grounded assessment of machine learning claims in drug discovery [36]. First, the panel is restricted to antiparasitic activity on this frozen dataset; cellular phenotype profiles (LISH/MoA) involve transcriptomic readouts that fingerprint locality and topological invariants do not govern, and the molecular–phenotypic boundary is explicitly maintained throughout (Section S1). Second, early-recognition metrics are degenerate on the present panels by construction: at $\approx 74\%$ active prevalence, EF@1 % saturates at 1.32–1.35 against a prevalence-imposed ceiling of ≈ 1.35 (precision@1 % 0.978–0.999), so no virtual-screening precision claim — including for the triage filter — is supported on this panel, and prospective evaluation on realistic low-hit-rate libraries is explicitly out of scope and reserved for future work (Section S6). Third, the GIN–TFP cohort gains are panel-bounded and descriptive: they are small (approximately 0.010–0.014 AUC points), the recomputed molecule-level intervals exclude zero outside the small polycyclic cohort ($n = 363$), and the aggregate scaffold-split gain remains modest (0.0091 points); no prospective or mechanistic claim follows. Results should be read as bounded to this panel and the distribution-shift settings studied here; replication on other targets, label distributions, and libraries is required before broader generalisation.

A constraint that is often treated as peripheral is here central: the entire pipeline must be deployable without commercial cloud infrastructure. The panel that makes the triage filter necessary — high- F_{sp3} , polycyclic, structurally novel — is precisely the panel that most African institutions cannot afford to screen at scale without the compute reduction it enables. The centroid-based screening reduction (263 000–1936 representative evaluations, 99.3 % overhead reduction) is therefore not a convenience optimisation; it is the condition under which autonomous drug discovery capacity exists in malaria-endemic African research institutions. The centroid selection preserves structural diversity (mean pairwise Tanimoto similarity between centroids: 0.32) and covers 94.7 % of the chemical space coverage (measured by ACSI retention), ensuring representativeness without information loss.

4 Conclusions

Under chemical distribution shift on this panel, model choice in virtual screening is not a global decision — it is a spatial, distance-dependent one. The distance-resolved benchmark shows that

ECFP4–RF’s largest advantages occur in the structural extrapolation tail ($D_{\text{NN}} < 0.40$), that an interpretation consistent with the 1-Weisfeiler–Lehman (1-WL) expressivity limitation of message-passing networks accounts for the observed pattern on high- F_{sp3} polycyclic scaffolds, and that multiscale topological fusion yields a small, replication-consistent gain over base GIN in every structural cohort (the interval excludes zero except on the small polycyclic cohort).

Two methodological contributions emerge from this work. The Distance-Aware Conformal Triage Filter formalises distance-dependent routing as a testable operational hypothesis for settings where model superiority is spatially heterogeneous, with conformal coverage near the nominal 90 % level on the five routed arms and no prospective precision gain claimed. These boundaries align with recent work on robust molecular-property evaluation [37, 38] and on uncertainty quantification under data shift [39]. Fold-independent weight restoration in ChemBERTa reduces the risk of cross-fold information transfer in transformer benchmarks — a reproducibility safeguard, not an optimisation. The restoration protocol complements ongoing community efforts toward open, reproducible chemical foundation-model training [40].

The 99.3 % compute reduction is not a performance footnote — it is the condition under which this framework is usable by the institutions that need it most. Centroid-based screening (263 000 to 1936 evaluations) brings multi-target virtual screening of African natural product libraries within reach of HPC infrastructure in malaria-endemic research institutions, without dependence on commercial cloud resources. What remains is biology: triage-selected candidates must now face drug-resistant *Plasmodium falciparum* isolates in vitro — the only test that converts a ranked list into an antimalarial lead.

5 Methods

5.1 Data curation and structural descriptors

We evaluate representations on a frozen panel of 19 836 antimalarial molecules curated from the eos80ch collection [9], originating from African Natural Product Database (ANPDB) seeds and synthetic analogues. Molecular representations comprise 2048-bit ECFP4 Morgan fingerprints (radius 2) [41], 78-dimensional topological feature projections (TFP), and 192-dimensional tensor-network embeddings (TNE). Structural complexity metrics, including the fraction of sp^3 -hybridised carbons (F_{sp3}), ring architecture counts, and scaffold entropy, were computed using RDKit [42].

5.2 Partition families and statistical evaluation protocol

Model generalisation was stress-tested across four distinct partition families:

1. **Stratified random split.** Uniform 80/10/10 partitioning serving as the in-domain baseline.
2. **Canonical scaffold split.** Bemis–Murcko ring-core separation [43] enforcing structural extrapolation.
3. **Butina cluster split.** Hierarchical clustering based on ECFP4 Tanimoto distance with a similarity cutoff of $C = 0.55$.
4. **NN-Tanimoto decile stratification.** Test candidate partition into deciles D1–D10 ordered by nearest-neighbour similarity to the training set.

All evaluation protocols adhere to a repeated 5×5 cross-validation design (25 fold–seed execution records per model architecture and partition scheme). Paired hypothesis testing operates on the five per-seed performance means ($df = 4$) with Benjamini–Hochberg false discovery rate adjustment; exact adjusted p-values are reported per comparison (e.g. p_{adj} 0.0002–0.0378 for the scaffold-split contrasts in Table 1), not summarised by a single tail bound.

5.3 Model architectures and leakage-risk-reducing transformer initialisation

Five primary architectures were benchmarked:

- **ECFP4–RF.** Random Forest classifier (500 trees) trained on 2048-bit ECFP4 fingerprints.
- **GIN.** 3-layer Graph Isomorphism Network (128 hidden units, dropout $p = 0.1$, ReLU activations, batch normalisation after each message-passing layer), bounded by 1-Weisfeiler–Lehman (1-WL) expressivity limits.
- **GIN–TFP.** Multiscale hybrid architecture concatenating the GIN graph embedding with a 78-dimensional persistent-homology feature vector (H_0 connectivity persistence: dims 0–32; H_1 cycle persistence: dims 33–57; scalar Betti invariants: dims 58–77) prior to the classification head.
- **GIN–TNE.** Hybrid architecture replacing the TFP block with a 192-dimensional Tucker-decomposed tensor-network embedding.
- **ChemBERTa.** Pretrained sequence transformer (ChemBERTa [44]; `seyonec/ChemBERTa-zinc-base-v1`) fine-tuned on canonical SMILES strings. To reduce the risk of cross-fold information transfer, pretrained weights were restored from the original checkpoint before each cross-validation fold, so that no fold’s fine-tuned weights seed another fold — a leakage-risk-reducing protocol for valid transformer benchmarks.

5.4 Distance-aware conformal triage formulation

The Distance-Aware Conformal Triage Filter evaluates a query candidate molecule x against training library $\mathcal{T}$ by calculating its nearest-neighbour Tanimoto similarity $D_{\text{NN}}(x) = \max_{y \in \mathcal{T}} s_{\text{Tanimoto}}(x, y)$ and batch Expected Calibration Error $\text{ECE}(x)$. The $\text{ECE} > 0.08$ threshold was calibrated using a validation curve on held-out predictions, balancing false positive and false negative rates for calibration failures (see [Section S6](#)). Candidates are routed according to the decision rule:

$$\mathcal{M}(x) = \begin{cases} \text{ECFP4–RF}, & D_{\text{NN}}(x) < 0.40 \quad (\text{Cold OOD Extrapolation Tail}), \\ \text{GIN–TFP}, & 0.40 \leq D_{\text{NN}}(x) \leq 0.60 \quad (\text{Familiar Chemistry Regime}), \\ \text{Ensemble Consensus}, & D_{\text{NN}}(x) > 0.60 \quad (\text{High-Similarity Core}). \end{cases} \quad (1)$$

When batch confidence degradation breaches the threshold $\text{ECE} > 0.08$, post-hoc Platt scaling or isotonic regression is triggered prior to candidate ranking. A Python specification of the decision rule is provided in Supporting Information ([Section S7](#)); its thresholds were informed by the calibration and decile analyses of this panel, and the interface is reproducible research code rather than validated deployment software.

Declarations

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Conflicts of interest: There are no conflicts to declare.

Ethics statement: This study is entirely computational. No human participants, animal subjects, or biological samples were involved.

Author contributions (CRediT): M.V.S.T.: Conceptualisation, Methodology, Software, Formal Analysis, Writing – Original Draft. J.-P.T.N.: Data Curation, Writing – Review & Editing. P.S.: Writing – Review & Editing. W.F.M.: Supervision, Methodology – Review & Editing. S.G.N.E.: Conceptualisation, Methodology, Supervision, Project Administration, Writing – Review & Editing.

Data availability: All dataset split files, prediction outputs, and analysis scripts are archived on

Zenodo (DOI: <https://doi.org/10.5281/zenodo.22787439>). The Zenodo package is reserved pending final upload; editors and referees will be granted immediate access to the full dataset and code for peer review via a shared link, and the DOI will be published upon acceptance. All models, training scripts, and benchmark configurations are publicly available to ensure reproducibility and transparent validation.

Computational resources: Experiments were conducted on the penavoraserver HPC cluster, using Python 3.11.15, PyTorch 2.13.0+cu130, PyTorch Geometric 2.8.0, transformers 5.14.1, RDKit 2025.03.6, and scikit-learn 1.9.0 (versions locked in the repository’s `requirements.txt`).

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